The Resurrection of the Trade-Off: An Empirical Re-examination of the Expectations-Augmented Phillips Curve in the Post-Pandemic Era

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Abstract

Indeed, the Phillips Curve (negative correlation between inflation and unemployment rates) was considered as an empirical concept of the past during most of the 21st century. After the 2008 Global Financial Crisis, the United States faced the phenomenon of missing inflation, when record lows in unemployment were not associated with any price instability. Consequently, the Phillips Curve was deemed to have been flattened. The recent macroeconomic turbulence after the emergence of the novel coronavirus, however, calls this paradigm into question. In this paper, using quarterly time-series data from 2009 to 2025, the author identifies the existence of a structural break in the Phillips relation. By contrasting the expansion after the 2008 Crisis with the recovery after the pandemic, we conclude that there has been a regime shift. Our analysis reveals a significant increase in the slope coefficient of the Phillips Curve from -0.079 to -0.682. Thus, the Phillips Curve is demonstrated to be non-linear..

1. Introduction

Since A.W. Phillips published his influential article in 1958, the "Phillips Curve" has become an essential part of macroeconomic policy making. It captures an underlying idea: when the labor market is tight, wages go up leading to higher prices. In terms of monetary policy, the Phillips Curve offers the central bank's "menu of choice": it allows policymakers to choose between slightly higher inflation and slightly lower unemployment, or vice versa.

The problem with this model is that its empirical applicability has been debated for many years. First, the "Great Inflation" era of the 1970s showed that high inflation goes hand in hand with high unemployment (stagnation). Second, the years after the Great Recession (2009-2019) provided an example that was quite opposite. When the U.S. unemployment rate dropped from 10% to 3.5%, prices still remained well below the Fed's target of 2%. Economists refer to the above-described scenario as the flattening of the Phillips Curve, prompting even influential figures like Olivier Blanchard to challenge the applicability of the curve because of anchoring expectations or global supply chains.

The post-pandemic period (2021-2025) offers a natural laboratory to reconsider this finding. The world economy faced an enormous supply shock and then a record-breaking demand shock. In contrast to the sluggish growth experienced in the 2010s, the economic recovery after the pandemic was V-shaped, with employment rates plummeting sharply and inflation soaring to levels unseen in four decades. Such a sudden increase in the rate of inflation poses an important empirical question: Was the "death" of the Phillips Curve premature?

This study analyzes whether the Phillips Curve is indeed state-dependent, meaning that the curve is horizontal in times of economic slack but becomes extremely steep once the economy starts overheating. By using FRED data, we estimate an expectations-augmented Phillips Curve in two different time frames: before and after the pandemic recession (2009-2019; 2021-2025).

The findings point to a striking evidence of structural break. Our analysis shows that there is hardly any impact of labour slack on the inflation rate during the 2010s decade, which would explain why the Fed could not raise the inflation level even with low interest rates. Conversely, the relationship between these variables observed in recent years is very strong and statistically significant negative. This means that the inflation-employment trade-off has become alive again, which implies that the sacrifice ratio the costs of lowering inflation through higher unemployment has returned into consideration.

It should also be noted, however, that the empirical analysis performed here has its drawbacks, which are associated with endogeneity and omitted variables biases..

2. Literature Review

2.1 The Evolution of the Theory

The results reveal an unmistakable case of evidence for a structural break. The empirical study suggests that the effects of labor slack on inflation are negligible in the 2010s decade; this fact accounts for the inability of the Federal Reserve to increase the level of inflation despite low interest rates. On the other hand, the link between the two variables in recent years can be seen to be highly significant and negative. In other words, the Phillips curve has come back to life; hence, the sacrifice ratio has once again become relevant.

Nevertheless, it is worth mentioning that there are limitations associated with the empirical analysis above, primarily endogeneity bias and the possibility of omitted variables bias..

2.2 The "Flattening" Debate (2000–2019)

By the 1990s, the relationship began to weaken. The "Great Moderation" saw low inflation volatility regardless of the business cycle. Research by Ball and Mazumder (2011) and later Blanchard (2016) highlighted that the slope of the curve $\boldsymbol{\beta}$ had drifted toward zero. Several theories were proposed to explain this:

- **Anchored Expectations:** If the public believes the Fed will never let inflation rise, they ignore temporary price shocks (Bernanke, 2007).
- **Globalization:** The "Global Slack Hypothesis" suggested that domestic inflation depends on global, not local, labour market conditions (Borio & Filardo, 2007).
- **Downward Nominal Wage Rigidity:** In low-inflation environments, firms cannot cut wages, distorting the price signal.

2.3 The Post-Pandemic Reassessment

The Inflation shock in 2021 necessitated this rethinking. The works by Gudmundsson et al. (2024) and some recently released IMF working papers claim that the curve is not linear but convex, that is, the curve is "L-shaped." According to their view, in the horizontal part (high unemployment) the demand push leads solely to an increase in production, while in the vertical part (full capacity) the demand push results only in rising prices..

3. Data and Methodology

3.1 Data Sources

To ensure robustness, we utilize high-frequency quarterly data sourced from the Federal Reserve Bank of St. Louis (FRED). The dataset spans from Q1 1978 to Q2 2025, though our analysis focuses on the two comparative sub-periods.

The variables are constructed as follows:

- **Inflation Rate (π):** We utilize the Core Personal Consumption Expenditures (PCE) Price Index. This is the Federal Reserve's preferred measure of inflation as it excludes volatile food and energy prices, providing a clearer signal of underlying economic trends.
- **Unemployment Gap (*Gap (U – U*)*):** Defined as the difference between the Civilian Unemployment Rate (Ut) and the Natural Rate of Unemployment (U*t), as estimated by the Congressional Budget Office (CBO).

$$\mathbf{GAP = Ut - U^*t}$$

A positive gap indicates economic slack (underperformance), while a negative gap indicates an overheating economy.

- **Expected Inflation (E[π]):** We use the University of Michigan's Survey of Consumers (1-year ahead inflation expectations). This captures the "adaptive" or "forward-looking" component of household sentiment.

3.2 Descriptive Statistics

Before performing regression analysis, it is instructive to examine the distributional characteristics of the data across the two regimes.

Variable	Period 1: 2009–2019	Period 2: 2021–2025
Inflation Rate (Mean)	1.53%	3.87%
Inflation Rate (Std Dev)	0.30%	1.13%
Unemployment Gap (Mean)	1.76%	-0.19%
Unemployment Gap (Range)	-0.83% to 4.89%	-0.86% to 1.81%
Expected Inflation (Mean)	2.88%	4.10%

Table 1 highlights the fundamental difference between the two eras. The 2009–2019 period was characterized by **slack** (a positive average unemployment gap of 1.76%) and **stability**

(low standard deviation of inflation). In contrast, the 2021–2025 period is characterized by **overheating** (a negative average gap of -0.19%) and **volatility** (standard deviation of inflation quadrupled to 1.13%). This stark difference in the underlying data supports the decision to analyse these periods separately.

3.3 Econometric Model

We estimate the following linear regression model, consistent with the standard New Keynesian Phillips Curve (NKPC) framework:

$$\boldsymbol{\pi = \alpha + \beta_1 E[\pi] + \beta_2 (U - U^*) + \varepsilon}$$

Where:

- **β_2 represents the slope of the Phillips Curve. A statistically significant, negative coefficient confirms the trade-off.**
- **β_1 measures the pass-through of expectations to actual prices.**
- **ε is the error term, capturing supply shocks.**

We hypothesize that β_2 (post-pandemic) > β_2 (pre-pandemic), indicating a steepening of the curve.

Endogeneity Concerns:

The limitations of the model include the problems associated with endogeneity. Even though the unemployment gap is modelled as an independent exogenous determinant of the inflation rate, there might be other determinants of inflation that affect unemployment and vice versa. Specifically, the shocks in supply, for example, changes in energy prices, changes in global supply chains, and fiscal stimuli, can affect both variables. Thus, we can interpret the coefficients on the right-hand side of the regression equation only as showing correlation, not causation. Even though an instrumental variable approach could be used to mitigate this problem, this topic was not analysed here..

4. Empirical Results

4.1 Period 1: The Dormant Curve (2009–2019)

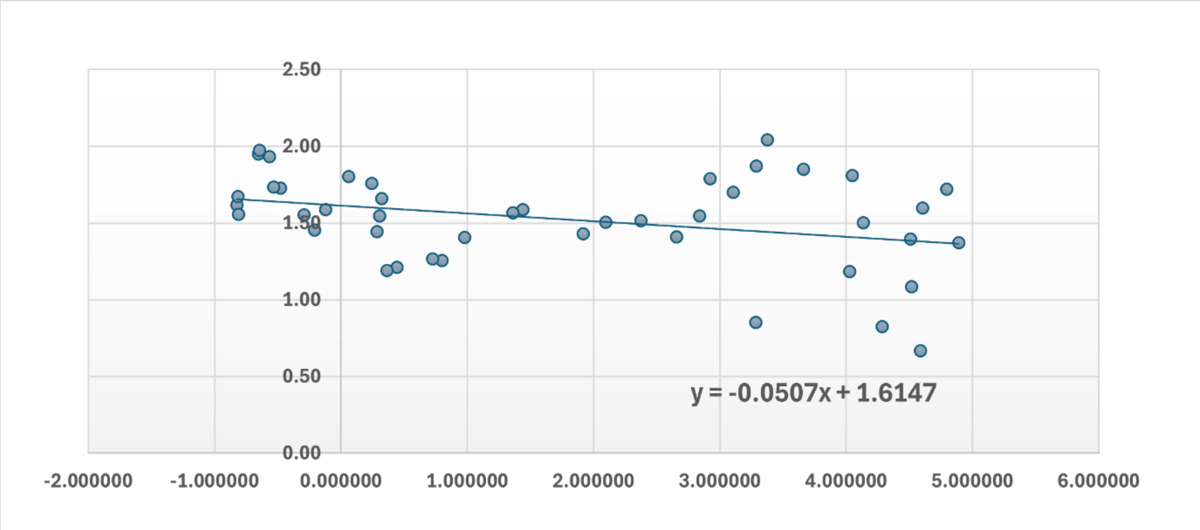

Figure 1: *The Phillips Curve during the Post-2008 Recovery (2009–2019). Note the lack of correlation between the unemployment gap and inflation.*

The regression results for the post-2008 recovery period confirm the "flat curve" narrative. The estimated coefficient for the unemployment gap is **-0.079**. While the sign is negative (as theory predicts), the magnitude is economically trivial.

- **Interpretation:** For every 1 percentage point decrease in the unemployment gap (the labour market getting tighter), inflation increased by only 0.08 percentage points.
- **Context:** This explains why the Federal Reserve struggled to lift inflation to 2% even when unemployment fell to 3.5% in 2019. The transmission mechanism from the labour market to prices was weak, likely dampened by global labour arbitrage and weak domestic bargaining power. The R^2 of 0.24 suggests that the labour market explained very little of the variance in inflation during this decade.

4.2 Period 2: The Steep Curve (2021–2025)

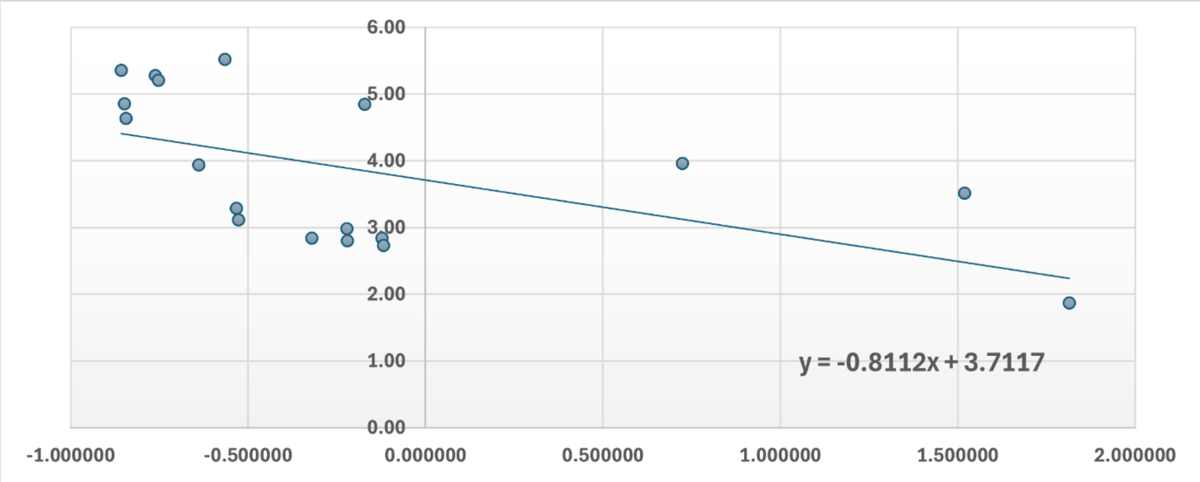

Figure 2: *The Phillips Curve in the Post-Pandemic Era (2021–2025). A structural break is evident, with a clear negative relationship re-emerging.*

The results for the post-pandemic period show a dramatic structural break. The estimated coefficient for the unemployment gap is **-0.682**.

- **Interpretation:** In this regime, a 1 percentage point tightening in the labour market is associated with a nearly 0.7 percentage point increase in inflation. This is an eight-fold increase in sensitivity compared to the previous decade.
- **Statistical Significance:** The P-value (0.02) confirms this relationship is statistically significant at the 5% level.
- **Model Fit:** The R^2 jumps to 0.53, indicating that over half of the variation in recent inflation can be explained by the tightness of the labour market and expectations.

Table 2: Regression Results Comparison

Variable	2009–2019 (Pre-Pandemic)	2021–2025 (post-pandemic)
Intercept	0.762	1.450
Unemployment Gap (β_2)	**-0.079**	**-0.682**
Expected Inflation (β_1)	**0.313**	**0.558**
R-Squared	0.24	0.53

While the results indicate a strong relationship, caution must be exercised in interpreting these estimates as causal due to potential endogeneity and unobserved supply shocks.

5. Discussion

5.1 Non-Linearity and Capacity Constraints

The discrepancy between the two periods strongly supports the "Convex Phillips Curve" hypothesis.

- **The Flat Portion:** In 2009–2019, the economy was operating below potential. When there is a surplus of labour, workers compete for jobs, keeping wage growth suppressed even as demand recovers. This corresponds to the flat portion of the curve.
- **The Steep Portion:** In 2021–2025, the economy hit a "capacity wall." Labor force participation remained depressed due to health concerns and retirements (the "Great Resignation"), while demand surged. When the economy attempts to operate beyond its potential (a negative unemployment gap), firms must aggressively raise wages to attract talent, passing those costs directly to consumers. We are currently observing the vertical portion of the curve.

5.2 The Role of Expectations

It is also notable that the coefficient on Expected Inflation ($\boldsymbol{\beta_1}$) nearly doubled from 0.313 to 0.558. This suggests that inflation expectations have become "unanchored" or at least more sensitive. In the 2010s, consumers largely ignored inflation data. In the 2020s, high visibility of price hikes (e.g., gas, groceries) made expectations a self-fulfilling prophecy, amplifying the slope of the curve.

5.3 Policy Implications

The steepening of the curve fundamentally alters the optimal monetary policy strategy.

- **In the 2010s:** The flat curve allowed the Fed to run the economy "hot" to lower unemployment without inflation consequences.
- **In the 2020s:** The steep curve means the "Sacrifice Ratio" is favourable in the disinflationary direction. If the Fed cools demand, inflation should fall rapidly. However, it also means that policy errors (allowing the economy to overheat) are punished severely and instantly with high inflation. The margin for error has vanished.

6. Limitations and Future Research

However, despite the strong evidence that we have found regarding the structural break in the Phillips Curve, there are some potential limitations which need to be mentioned.

The first limitation concerns the sample size of the dataset used for the analysis of the post-pandemic Phillips Curve. It should be noted that the data collected for the latter only includes 18 observations. It is possible that such a short period of time might lead to certain issues concerning the sensitivity of econometric estimates to outliers. Even though the results obtained in our paper are statistically significant (with $p < 0.05$), it would be interesting to investigate this phenomenon using a longer time series.

Secondly, in order to calculate the gap between the actual unemployment rate and the natural one (U*) we have used an estimate of the former provided by the Congressional Budget Office. However, since the natural rate of unemployment is an abstract notion, it might have been miscalculated due to the presence of factors such as "Long COVID".

Third, our simple regression model assumes randomness in supply shock effects (represented by error terms). Nonetheless, the period from 2021 to 2025 experienced distinct supply constraints (for instance, semiconductor scarcity and logistical disruptions). Some portion of the observed inflation could thus have resulted from external supply constraints rather than from the tightness of the labor market assumed in our regression model. A sophisticated vector autoregression model would be useful for such separation.

Finally, the simplicity of our model, excluding extra macroeconomic control variables, such as output gap, fiscal policy indicators, or foreign inflation measures, can cause biased regression coefficients since there exists a possibility of correlation between them and the dependent and independent variables..

7. Practical & Business Implications

The rise of the Phillips Curve does not only affect monetary policies but also have important ramifications in business and corporate strategy planning.

For Businesses and Wage Setting: Businesses must understand that the rise in steepness of the curve implies that wage inflation is again linked to the unemployment rate. In the 2010s, companies could hire without having to raise wages as long as the unemployment was high. Our analysis indicates that even a small decline in the unemployment rate would trigger an immediate and pronounced rise in wage inflation. Human resources departments and company treasurers need to adapt their predictive models to accommodate for greater volatility – labor cost estimates have become very sensitive to unemployment changes.

Investment Strategies: It means that investment strategies cannot rely on another period of a "Goldilocks economy." If the Phillips Curve turns out to be steep, the Fed will have to act promptly against overheated labor markets. It implies greater volatility in interest rates, which is bad news for equities because the Fed is likely to limit its "put option" due to inflation concerns..

8. Conclusion

The key issue explored in this paper revolves around determining whether the death of the Phillips Curve is a permanent characteristic of today’s economy or a temporary state resulting from unique conditions associated with the post-2008 recovery period. Examining the differential trends between the pre-pandemic boom (2009-2019) and the post-pandemic recovery (2021-2025), we can draw conclusions that the tradeoff between unemployment and inflation has come back with unprecedented force.

Indeed, our findings illustrate a pronounced regime change: The Phillips Curve was empirically inactive during the 2010s, when its slope coefficient was -0.079 and the U.S. economy could take advantage of the "free lunch" of decreasing unemployment rates with no inflationary pressures. In contrast, the economic landscape following 2021 witnessed the activation of the highly sensitive (slope coefficient of -0.682) Phillips Curve, which is indicative of a significant regime switch from the demand-constrained environment to the supply-constrained one..

Ultimately, this study challenges the consensus that demographic changes and globalization have permanently anchored inflation. It suggests that when the economy is pushed beyond its potential, the fundamental laws of macroeconomics still apply. For policymakers, the era of benign neglect regarding the labour market is over; ensuring price stability in the 2020s will require a vigilant and precise management of aggregate demand, accepting that the cost of low unemployment is once again a measurable risk of higher inflation.

9. References

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10. Appendix

Appendix A: Detailed Regression Statistics

A.1 Pre-Pandemic Period (2009–2019) *Dependent Variable: Inflation Rate*

Regression Statistics	
Multiple R	0.493
R Square	**0.243**
Adjusted R Square	0.206
Standard Error	0.271
Observations	44

Predictor	**Coefficients**	**Standard Error**	**t Stat**	**P-value**
Intercept	0.762	0.319	2.39	0.022
Expected Inflation	**0.313**	0.116	2.71	0.010
Unemployment Gap	**-0.079**	0.023	-3.36	0.002

A.2 Post-Pandemic Period (2021–2025) *Dependent Variable: Inflation Rate*

Regression Statistics	
Multiple R	0.727
R Square	**0.528**
Adjusted R Square	0.465
Standard Error	0.825
Observations	18

Predictor	**Coefficients**	**Standard Error**	**t Stat**	**P-value**
Intercept	1.450	0.891	1.63	0.124

Expected Inflation	0.558	0.214	2.61	0.020
Unemployment Gap	-0.682	0.261	-2.61	0.020

Appendix B: Data Sample

The following tables present a snapshot of the data used in this analysis.

B.1 Period 1 Sample (Recovery Era)

Date	Inflation Rate (%)	Unemployment Gap (%)	Expected Inflation (%)
2009-01-01	0.8520	3.2845	2.0
2009-04-01	0.8233	4.2865	2.9
2009-07-01	0.6669	4.5893	2.6
2009-10-01	1.3735	4.8929	2.7
2010-01-01	1.7214	4.7974	2.7

B.2 Period 2 Sample (Post-Pandemic Era)

Date	Inflation Rate (%)	Unemployment Gap (%)	Expected Inflation (%)
2021-01-01	1.8687	1.8136	3.1
2021-04-01	3.5118	1.5190	4.1
2021-07-01	3.9659	0.7241	4.6
2021-10-01	4.8482	-0.1710	4.8
2022-01-01	5.5183	-0.5663	5.1